

TTT-01 • BIOLOGY + ENERGY

Mud Battery Bioelectric League

*Can your team make mud generate electricity and
defend your design with data?*

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN

Highest combined score from peak reading, daily data quality, controlled-variable design, redesign notes, and final explanation.

TTT-02 • FIELD SCIENCE

Forest Sensor Sprint

Find the smartest place to install a future campus tree sensor.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN

Best evidence-based sensor recommendation with accurate measurements and efficient route planning.

TTT-03 • BIOMIMICRY

Seed Dispersal Derby

Build a seed that escapes the parent tree without engines or throwing.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN

Best combined distance, hang time, target landing, biomimicry explanation, and redesign improvement.

TTT-04 • BIOMIMICRY

Xylem Pipeline Relay

*Move water uphill like a tree, faster and cleaner
than rival teams.*

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN

Most water delivered with speed, low spill, and strong
explanation.

TTT-05 • ECOLOGY

Greenhouse Climate Controller

Run the greenhouse controls for a plant that cannot complain, only wilt.

PREDICT › BUILD or SOLVE › TEST › SCORE › REDESIGN

WIN

Best plant-zone-control match supported by tour evidence.

TTT-06 • BIOMIMICRY

Pinecone Weather Machine

Build a humidity sensor with no battery and no code.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN

Most accurate, readable, and fast response after spray-and-dry trials.

TTT-07 • ECOLOGY

Pollinator Network Draft and Build

*Draft a pollinator habitat that works all season,
not just on the prettiest day.*

PREDICT › BUILD or SOLVE › TEST › SCORE › REDESIGN

WIN

Highest pollinator support across spring, summer, and fall using
native and clumping logic.

TTT-08 • FIELD SCIENCE

Arboretum Eco-Quest

Read the living collection and solve the checkpoints before time runs out.

OBSERVE › PLAN ROUTE › VERIFY › RECORD EVIDENCE › SCORE

WIN

Most accurate checkpoint solutions with safe field conduct.

TTT-09 • ECOLOGY

Minecraft Tree World Resilience Cup

*Design a landscape that survives stress because of
how it is built, not luck.*

PREDICT › BUILD or SOLVE › TEST › SCORE › REDESIGN

WIN

Best climate-resilient design with accurate nature-based
strategies.

TTT-10 • DATA + CLIMATE

Tree Ring Climate Detective

Use an authored ring-card code to build and defend a model history, then explain what real evidence would require.

OBSERVE › APPLY CARD CODE › CITE BANDS › CHECK LIMITS ›
SCORE

WIN

Most accurate use of the practice-card code, supported with claim-evidence-reasoning.

TTT-11 • MATERIALS

Lotus Leaf Surface Sprint

*Build a surface that sheds water and dirt the way
a lotus leaf does.*

PREDICT › BUILD or SOLVE › TEST › SCORE › REDESIGN

WIN

Best clean runoff and least residue after one redesign.

TTT-12 • FIELD SCIENCE

Leaf Stomata Microscope Detective

Compare stomatal density with a standardized count and make a cautious, testable hypothesis.

STANDARDIZE › COUNT 3+ FIELDS › MEAN + RANGE ›
HYPOTHESIZE › SCORE

WIN

Most reliable density comparison with mean, range, and an evidence-limited hypothesis.

TTB-01 • FIELD SCIENCE

Root Grip Erosion Rescue

Can roots hold a slope together during a storm?

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN

Most soil retained after standard pour.

TTB-02 • FIELD SCIENCE

Tree Height Triangulation Shootout

Measure a tree without climbing it.

PREDICT › BUILD or SOLVE › TEST › SCORE › REDESIGN

WIN

Closest to instructor reference height.

TTB-03 • DATA + CLIMATE

Urban Heat Shade Dash

Find the coolest route through a hot campus.

PREDICT › BUILD or SOLVE › TEST › SCORE › REDESIGN

WIN

Best data-backed cool route.

TTB-04 • FIELD SCIENCE

Photosynthesis Float-Off Playoffs

Make leaf disks float by producing oxygen.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN

Fastest half-float time with controlled variables.